Tutorial-3

S.no	Floor Area (sqm) (x_1)	Distance from City Center (km) (x_2)	Bedrooms (x_3)	Building Age (years) (x_4)	Monthly Rent (₹) (y)
1	45	2.0	1	5	18,000
2	60	3.5	2	8	24,000
3	75	5.0	2	10	28,000
4	90	6.0	3	12	32,000
5	55	4.0	1	6	21,000

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

$b_1, b_2, b_3, b_4 \Rightarrow$ weights
$x_1, x_2, x_3, x_4 \Rightarrow$ input variables
$y \Rightarrow$ target y

MSE calculation $= \frac{1}{n}\sum_{i=1}^{n}(y_i - y_{pred})^2$

$\Rightarrow$ Initializing weights $\Rightarrow$ $b_0 = 0$, $b_1 = b_2 = b_3 = b_4 = 0$

$n = 5$

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 = 0 + 0x_1 + 0x_2 + 0x_3 + 0x_4$$

$$y = 0$$

$\Rightarrow \boxed{y_{pred} = 0}$

$$MSE = \frac{1}{5}\left((18-0)^2 + (24-0)^2 + (28-0)^2 + (32-0)^2 + (21-0)^2\right) \times (1000)^2$$

$$= \frac{3149}{5} \times (1000)^2 = 629.8(1000)^2 = 629800000$$

$\{\eta = 0.000001\}$

$$w_j^{new} = w_j^{old} + \eta \frac{\partial error}{\partial w_j}$$

$$\frac{\partial error}{\partial w_j} = \frac{2}{n} \sum_{i=1}^{n} (y_i - y_{pred}) x_{ij}$$

i	x_1	x_2	x_3	x_4	y
1	x_{11}	x_{12}	x_{13}	x_{14}	y_1
2	x_{21}	x_{22}	x_{23}	x_{24}	y_2
3	x_{31}	x_{32}	x_{33}	x_{34}	y_3
4	x_{41}	x_{42}	x_{43}	x_{44}	y_4
5	x_{51}	x_{52}	x_{53}	x_{54}	y_5
					y

⇒ for w_1 :-

$$\frac{\partial error}{\partial w_1} = \frac{2}{n} \sum_{i=1}^{n} (y_i - y_{pred}) x_{i1}$$

$$= \frac{2}{5} \left((18-0)\cdot 45 + (24-0)60 + (28-0)75 + (32-0)90 + (21-0)55 \right) \times 1000$$

$$= \frac{2}{5} (8385)^{\times 1000} = 3354000$$

$$w_1^{new} = w_1^{old} + \eta \frac{\partial error}{\partial w_1} = 0 + (0.000001)(3354000)$$

$$w_1^{new} = 3.354$$

⇒ for w_2 :-

$$\frac{\partial error}{\partial w_2} = \frac{2}{5} \left((18-0)2 + (24-0)3.5 + (28-0)5 + (32-0)6 + (21-0)4 \right) \times 1000$$

$$= \frac{2}{5} (536000) = 214400$$

$$w_2^{new} = 3.354 + (0.000001)(214400) = 3.5684$$

⇒ for w_3:-

$$\frac{\partial error}{\partial w_3} = \frac{2}{5}\left((18-0)1 + (24-0)2 + (28-0)2 + (32-0)3 + (21-0)1\right) \times 1000$$

$$= \frac{2}{5}(239000) = 95600$$

$$w_3^{new} = 3.5684 + (0.000001)(95600)$$

$$= 3.664$$

⇒ for w_4:-

$$\frac{\partial error}{\partial w_4} = \frac{2}{5}\left((18-0)5 + (24-0)8 + (28-0)10 + (32-0)12 + (21-0)6\right) \times 1000$$

$$= \frac{2}{5}(1072000) = 428800$$

$$w_4^{new} = 3.664 + (0.000001)(428800) =$$

$$= 4.0928$$

⇒ $$\frac{\partial error}{\partial b_0} = \frac{-2}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

$$b_0^{new} = b_0^{old} - \eta \frac{\partial error}{\partial b_0}$$

$$\frac{\partial error}{\partial b_0} = \frac{-2}{5}\left((0-18) + (0-24) + (0-28) + (0-32) + (0-21)\right) \times 1000$$

$$= \frac{-2}{5}(-123000) = 49200$$

$$b_0^{new} = 0 - (0.0000001)\,49200$$

$$= -0.0492$$

$$\hat{y} = -0.0492 + 3.354x_1 + 3.5684x_2 + 4.0928x_4 + 3.664x_3$$

⇒ Data 1 (45, 2.0, 1, 5)

$$\hat{y}_1 = -0.0492 + 3.354(45) + 3.5684(2) + 3.664(1) + 4.0928(5)$$

$$= 182.14$$

⇒ Data 2 (60, 3.5, 2, 8)

$$\hat{y}_2 = -0.0492 + 3.354(60) + 3.5684(3.5) + 3.664(2) + 4.0928(8)$$

$$= 253.75$$

⇒ Data 3 (75, 5.0, 2, 10)

$$\hat{y}_3 = -0.0492 + 3.354(75) + 3.5684(5.0) + 3.664(2) + 4.0928(10)$$

$$= 317.59$$

→ Data4 (90, 6.0, 3, 12)

$$\hat{y}_4 = -0.0492 + 3.354(90) + 3.5684(6.0) + 3.664(3) + 4.0928(12)$$

$$= 383.32$$

→ Data5 (55, 4.0, 1, 6)

$$\hat{y}_5 = -0.0492 + 3.354(55) + 3.5684(4.0) + 3.664(1) + 4.0928(6)$$

$$= 226.91$$